

Scalability & Future Plan

Date	04 July 2026
Team ID	SWTID-2026-9794
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited dataset size	May reduce prediction accuracy for diverse applicant profiles.	High
2	Local deployment only	Application is accessible only on the local system, limiting scalability and remote access.	High
3	Prediction depends on available applicant details	Additional financial or credit history information could further improve prediction accuracy.	Medium

Scalability Plan :

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy the application on a cloud platform (Render, AWS, or Azure) to support multiple users simultaneously.
2	Data Storage	Uses CSV dataset for prediction	Store applicant information in MySQL or PostgreSQL for improved scalability and efficient data management.
3	Performance	Runs prediction on a single local machine	Optimize the machine learning model, improve preprocessing, and use caching for faster predictions.
4	Security	Basic Flask application	Implement HTTPS, secure input validation, user authentication, and encrypted communication in future

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve machine learning model accuracy using larger datasets, feature engineering, and hyperparameter tuning.	1–2 Months	Higher prediction accuracy and reliability.
Phase 2	Add more applicant attributes and improve prediction performance using advanced machine learning techniques.	3–4 Months	Better prediction quality and decision support.
Phase 3	Deploy the application on a cloud platform (Render, AWS, or Azure) for online access.	4–6 Months	Improved scalability, availability, and accessibility.
Phase 4	Develop a mobile-friendly version and an analytics dashboard for monitoring prediction statistics.	6–12 Months	Enhanced user experience and better reporting capabilities.